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Four-year comparison of water contents beneath a grass ley and a deciduous oak wood overlying Triassic sandstone in lowland England

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Summary Differences in the seasonal water dynamics of a sand soil overlying Triassic sandstone have been investigated to a depth of 9 m beneath both a grass ley and a pendunculate oak woodland within Clipstone Forest, Nottinghamshire, England. Fortnightly measurements with a neutron-probe over a four-year period allowed a comparison of the soil–rock water content to depths rarely reported. In spring, the rate of decrease of water content of the uppermost 2 m of soil was much greater under grass than oak woodland. In contrast, the rate of decrease under oak was greater after leafing out in May, while the rate of rewetting in early autumn was lower for this land-use until senescence and leaf-fall in late autumn. In the uppermost 2 m of soil – a depth that includes all plant roots – the soil moisture minima were between 47 and 58 mm lower under oak than grass in each year of monitoring. As a result of both these drier conditions and the comparatively late leaf fall of this species, penetration of the winter-season wetting front to 2 m was delayed by between one and three months at the oak site relative to grass in years of near-average rainfall. Rewetting at 9 m lags by 10–12 months, compared to the surface, giving an average penetration rate for the wetting front to this depth of 25–30 mm day^{-1}, with little observable differences between the land-uses. Preferential flow is evident under both sites, affecting the profile to 3 m in all years and to at least 6 m following winters experiencing exceptional rainfall.

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Introduction

Protracted deforestation means that the British Isles are amongst the least wooded parts of Europe (Environment Agency, 1998). England has been particularly affected, with only 8% of its territory under forest at the close of the 20th Century (Smith, 2002). Consequently, in 1995, the UK Government proposed a doubling of the area of woodland by the year 2045 (Her Majesty's Stationery Office, 1995). However, while the amenity, biodiversity and landscape benefits of this intention are undeniable, questions were raised concerning the possible impacts on water resources of such a large change in land-use (House of Commons Environment Committee, 1996). The negative impact of evergreen conifer afforestation on the quantity of runoff from upland British catchments that receive high annual rainfall had already been demonstrated conclusively (e.g. Law, 1956; Calder and Newson, 1979). Studies of the effects of deciduous broadleaved species on water-relations elsewhere in the world suggested that, relative to pasture, higher transpiration and interception reduces soil water recharge (e.g. Guevara-Escobar et al., 2000). However, available information about the impact of an intended increase in broadleaved deciduous woodland on components of the water-balance in drier areas of lowland Britain was both limited and contradictory. For example, studies by Harding et al. (1992) and Roberts et al. (2001) suggested that recharge beneath beech and ash woodland on shallow soils overlying Cretaceous chalk is similar to, or greater than, that under grass. Meanwhile, Finch (2000), in a comparison of mixed deciduous woodland and grassland overlying sands and gravels of the Tertiary Reading Formation, suggests the opposite within the same geographic province. Compounding the problem of understanding the effects of land-use change on the soil moisture regime was that there was only limited information available about evaporation from woodland growing on drought-prone sand soils overlying Triassic sandstone (Ragab et al., 1997) — the UK's second most important aquifer (Kinniburgh, 1999). Yet, much of the planned increase in woodland might take place in the English Midlands on this type of soil.

In light of this knowledge-deficit and given the ever-increasing pressures on groundwater resources to satisfy the needs of both irrigated agriculture and public water supply, the Department of the Environment, Transport and the Regions, supported by the Forestry Commission and the Environment Agency, commissioned a field study of water use by evergreen pine forest, deciduous oak woodland, grassland and heath at Clipstone Forest in Nottinghamshire, England (Fig. 1), of which data for the oak and grassland sites will be reported here. Clipstone Forest is part of the wider Greenwood Community Forest, within which it is planned to expand woodland from the current nine percent to 27% by the middle of the 21st century. The aim of the study was to assess the impact of different vegetation types on groundwater recharge of the Triassic Sherwood Sandstone through examination of water contents to a depth of 9 m. In support of this, soil and rock water content data were collected between February 1998 and April 2002. The depth of the soil—rock profile from which data were gathered is considerably greater than in most other studies (c.f. Bréda et al., 1995; Granier et al., 1999; Guevara-Escobar et al., 2000). The length of the record — four years — is also only rarely matched by other studies. It has allowed us to assess changes in the soil—rock water profile not only during periods of near normal amounts of rainfall, but also in years of comparatively unusual weather.

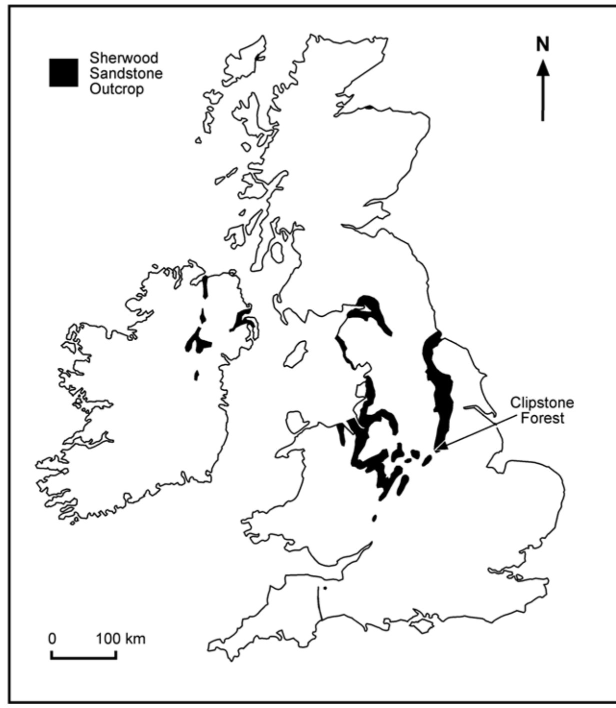

Figure 1 Outcrop of Triassic Sherwood Sandstone in the British Isles and the location of Clipstone Forest (centred upon UK grid reference SK 6162; 1°05′W, 53°09′N).

Experimental sites

The study was located within Clipstone Forest, a relatively large woodland in the midlands of England. Extending over an area of approximately 5×5 km, the forest is characterized predominantly by commercial plantations of Corsican pine (*Pinus nigra* var. *maritime*), but within it are sizeable tracts of deciduous woodland and heathland. All experimental sites were located within 2.5 km of each other on land sloping at <2° and at an altitude of 95 m above sea level.

Daily rainfall data from a non-recording UK Meteorological Office Mark II rain gauge at Gleadthorpe (UK Grid Ref: SK 593701; 1°07′W 53°13′N), 9 km north of the forest, provide a long-term (1971—2000) annual mean of 628 mm. Weekly estimates of potential evapotranspiration for grass, obtained from the UK Meteorological Office Rainfall and Evaporation Calculation System (MORECS), which uses a modified form of the Penman—Monteith equation (Meteorological Office, 1992), provide an equivalent long-term annual mean of 608 mm for the 40×40 km grid square within which Clipstone Forest lies. The small difference between annual rainfall and annual potential evapotranspiration (20 mm) raises concern over the sustainability of water resources where land-use changes might be instrumental in instigating significant shifts in water-balance.

Vegetation

A grass ley (predominantly rye grass, *Lolium perenne* L.) of approximately 3 hectares, located on the northern edge of the forest, was chosen to allow evaluation of soil—rock water redistribution under a common short arable crop that is not irrigated. Field management of the site included mowing twice a year — in June and August — and an annual application of manure or mineral fertilizer during spring. Prior to cutting, the grass sward was usually about 0.6 m in height. The over-wintering sward was typically less than 0.1 m high. The site was ploughed and reseeded once during the monitoring period, in the autumn of 1999/2000, though the immediate area of data capture was left untouched.

The oak woodland chosen for instrumentation is a forestry compartment of native, deciduous pedunculate oak (*Quercus robur* L.) planted in 1945 on heathland. At the time of field measurements, the average cross-sectional trunk area at breast height was 0.07 m^2 and tree density was 360 per hectare, giving a nearly closed canopy. Besides the tree canopy, there was a fairly well developed ground flora. This consisted primarily of bracken (*Pteridium aquilinum* L.), bramble (*Rubus fruticosus* agg.) and grasses. *Q. robur* develops its leaves late in spring compared with other native British trees, reaching 50% of the annual maximum by early to mid May. There is an equally late leaf fall, with half the leaves typically remaining in situ until mid to late November, though the leaves will usually start to turn colour at the end of September. Values of leaf-area index (LAI) were not established locally. However, the database compiled by Scurlock et al. (2001) for 38 sites containing *Quercus* spp. provides an average of 5.2 with a modest standard error and this is corroborated by studies such as that of Granier and Bréda (1996) for a stand of sessile oak (*Quercus petrea*), where LAI ranged between 4.2 and 6.0.

Observations made in deep soil pits indicate that most roots lay in the top 0.7—0.8 m at both sites. Below this, roots were found occasionally down to about 1 m at the grass site and down to 1.5—1.7 m at the oak wood. Below 1.8 m, the sandstone is unweathered and, therefore, indurated and well-cemented. This provides a natural barrier to root penetration. The vertical distribution of roots is similar to that revealed by Lucot and Bruckert (1992) for 150-year old *Quercus robur*. Here, despite being three times older than the oak at Clipstone, all classes of root lie predominately within 1 m of the surface and only 0.13% lies below 2 m, despite the fact that the 4-m deep colluvial soil provides no impediment to downward root extension. Bréda et al. (1995), reporting on a mixed stand of mature sessile and penduncculate oak growing in a deep loam, also show all classes of root to lie within 1.5 m of the surface.

While the oak site is situated 120 m from the forest edge, this edge is generally leeward and 74% of the run-of-wind has a fetch over woodland of at least 1 km. Neal et al. (1991) found that woodland edge effects on soil water relations are limited to a distance of <20 m. Given this and the pattern of fetch indicated by the wind-rose, the proximity of the woodland boundary is unlikely to have had a significant effect on water relations at the site of the measurements. Similar consideration of fetch was given to locating the instruments in the grassland and a site was chosen that was at least 200 m from the forest edge in the direction of the prevailing wind.

Soils

Soils at the sites are classified by the Soil Survey of England and Wales as either sandy podzols (oak: Crannymoor Series [Typic Haplorthod]) or brown earths (grass: Newport Series [Typic Quartzipsamment]) overlying Triassic Sherwood Sandstone (Robson and George, 1971; Ragg and Clayden, 1973). This classification, however, is based on widely distributed sampling. In reality, the soils at both sites are closely related and differences between the brown earths and podzols are very subtle, largely reflecting recent vegetation history and land management. The difference is greatest in the topsoil, which comprises a mull litter (0.05 m) and an A horizon (0.24 m) under the oak wood and an Ap horizon (0.25 m) under the grass ley. Data from a gamma probe suggests that soil density in the uppermost 0.5 m was slightly higher at the grass site than beneath the woodland, probably reflecting compaction associated with farm operations at the grass site and the incorporation of organic matter beneath the oaks. At both sites, the B/C horizon extends from around 0.25—0.29 to about 1.8 m and, to the eye, is identical under both land-uses. Below 1.8 m, the sandstone is essentially unweathered. Thin bands of marl occur sporadically in the soil—rock profile, but, typically, the soil material is 94% sand, 5% silt and 1% clay, and there is little deviation from this composition either between sites or with depth.

Both sites are free draining and show no signs of overland flow. The local groundwater table, as reflected by the height of local streams, lies at least 20 m below the soil surface, well beyond the rooting depth of the vegetation types under investigation. In fact, a borehole drilled to 36 m at one of the monitoring sites in March 2001 was found to be dry.

Instrumentation

Rainfall was recorded locally at Clipstone Forest using a Mk IV tipping-bucket rain gauge (nominal 0.2 mm per tip) attached to an automatic weather station and set in a broad clearing sited at least 100 m from the forest edge in all directions, and 700 m from the oak woodland. Regional values of potential evapotranspiration were derived by MORECS (Meteorological Office, 1992). Soil water content was monitored with a neutron probe (Didcot Soil Moisture Gauge type IH II) according to the method of Bell (1973). The probe was calibrated against volumetric gravimetric moisture content determinations in the local soil over a wide range of moisture content. Readings were taken at fortnightly intervals between February 1998 and April 2002, except for a three-month period (13th February—22nd May 2001) at the grass site due to a national epidemic of Foot and Mouth disease among livestock and a prohibition on access to farmland. Water content was measured in six access tubes at each site at 0.1 m intervals to a depth of 2 m, then at 0.2 m intervals to 4 m and every 0.5 m below this. The nominal depths of the tubes were: grass — 2.6, 2.6, 3.0, 6.0, 9.0 and 9.0 m; oak — 2.4, 3.4, 6.5, 7.5, 9.0 and 9.0 m.

The access tubes lay within 25 m of each other in the oak woodland, while, at the grass ley, they lay in two clusters of three tubes, 125 m apart. The tubes under oak were positioned in a stratified random manner to sample locations close to, and away from, tree trunks, in order to allow for likely spatial patterns of net rainfall, stemflow and preferential flow paths. The average standard error of the mean moisture content of all access tubes at each depth of measurement and on each measurement occasion was 1.4% and 0.6% at the grass and oak sites, respectively. The higher value for the grass ley probably reflects the wider spacing of the tubes and slight variability in soil properties.

Patterns of soil moisture redistribution

Seasonal extremes in soil—rock profile water content

The annual range in moisture content for each depth provides a measure of seasonal water uptake from the upper 2 m of soil during plant growth and an indication of the effect of drainage from the lower parts of the profile. Because there is a lag in reaching a state of either maximum or minimum moisture content with distance down profile, these states have been defined for each individual measurement depth and the data synthesized into wettest and driest profiles for each year (Fig. 2). Each profile has been constructed, therefore, with data derived from a number of measurement dates.

Several features in the profile are functionally related to textural characteristics of the soil—rock profile. For example, the persistent positive bulge in moisture content at a depth of 6 m under grass is almost certainly a product of a localised thin-layer of marl in the largely sandstone stratigraphy; the higher water content in the maxima curves at a depth of 1.5—2.0 m reflects the interface between the B/C soil horizon and the top of the indurated sandstone and a tendency to transient ponding of water as it moves down-profile; while the higher water contents in the curves for both minima and maxima in the top 0.5 m under oak are a function of the incorporation of organic matter in the A horizon under this land-use and its influence on water retention.

For each site, the synthetic minimum soil-moisture profiles are remarkably similar from year to year, particularly below a depth of 1 m. The four-year ranges in moisture volume fraction (dimensionless) at 0.5, 2, 5 and 9 m are 0.020, 0.008, 0.006 and 0.007 for grass and 0.013, 0.009, 0.010 and 0.004 for oak, respectively. The wettest profiles of each annual cycle also show a degree of consistency between years. The main exception was for oak in 2001—2002. As a result of the low autumn—winter rainfall and the prolonged period of interception that resulted from unusually late defoliation, rewetting in this instance had only reached a depth of 2.4 m by the end of April 2002 (A, Fig. 2), revealing the impact that unusual weather can have on water recharge and redistribution in these soils.

At a depth of 0.3 m, the difference between maximum and minimum values of moisture content (the averages of

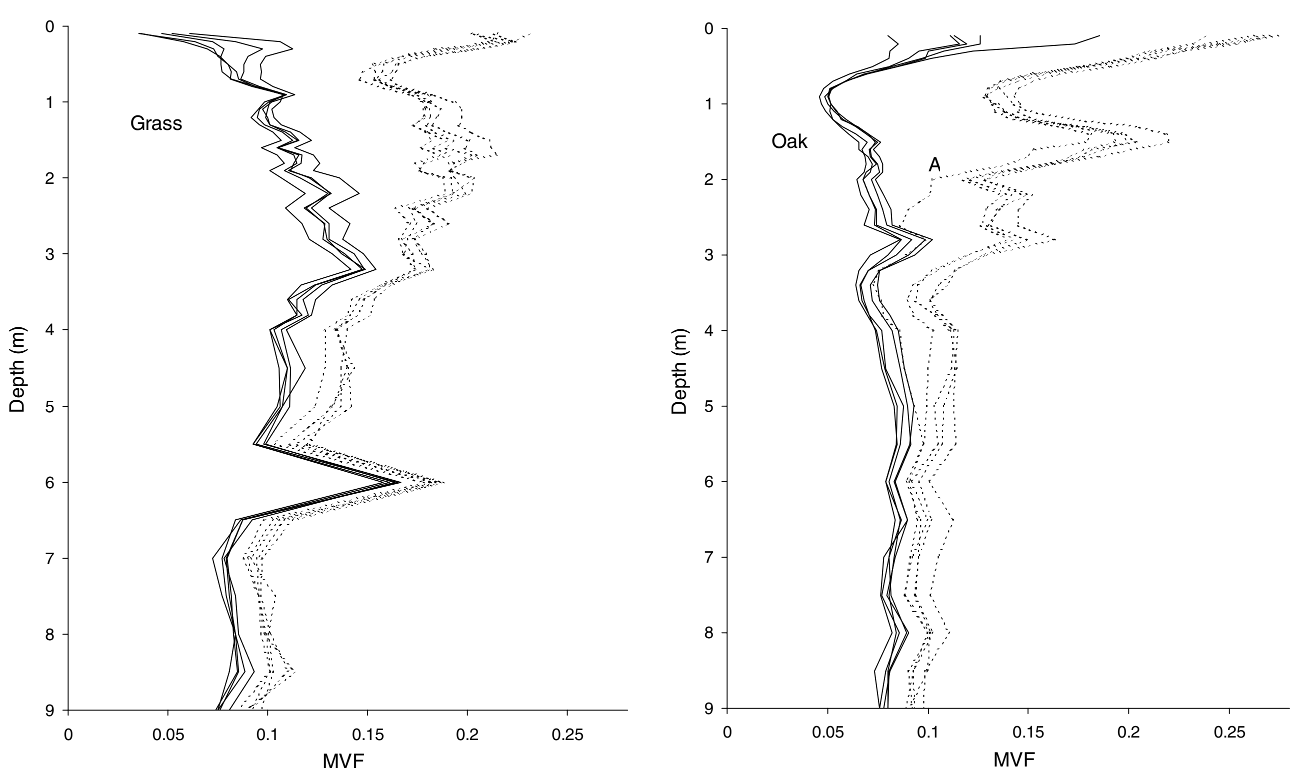

Figure 2 Synthetic annual minimum (1998—2001) and annual maximum (1998—2002) soil—rock moisture profiles (average of six access tubes in each case) under grass ley and oak woodland. Solid lines = minima; pecked lines = maxima; see text for explanation of annotation; MVF = moisture volume fraction (dimensionless).

all tubes at each site) over the four years was 0.136 and 0.157 for the grass and oak sites, respectively, declining to 0.047 and 0.057 at 3 m and only 0.025 and 0.022 at 9 m. This decrease in range with depth is expected, and has also been noted in chalk profiles by Mahmood-ul-Hassan and Gregory (2002). However, the ranges recorded at Clipstone exceed that reported in sandy soils under oak in Denmark, where winter and summer curves were effectively coincident at and below about 3 m (Ladekarl, 1998). Bréda et al. (1995) report variations in water content down to 2 m under a mixed stand of mature sessile and penduncluate oak growing on a deep loam, below which there appears to be no seasonal difference. Similarly, seasonal changes in moisture content were found to be virtually non-existent below a depth of 1.25 m in a heavy clay soil under grass in the UK (Reid and Parkinson, 1984). The results from Clipstone reported here are unique in highlighting the considerable depth of seasonal activity above and within the Triassic aquifer.

Soil water change with time and depth

Temporal changes in absolute moisture content are difficult to interpret due to the presence of occasional layers in the soil—rock profile that are consistently wetter or dryer as a function of subtle differences in texture or porosity. Consequently, the data for each site have been presented in rel-

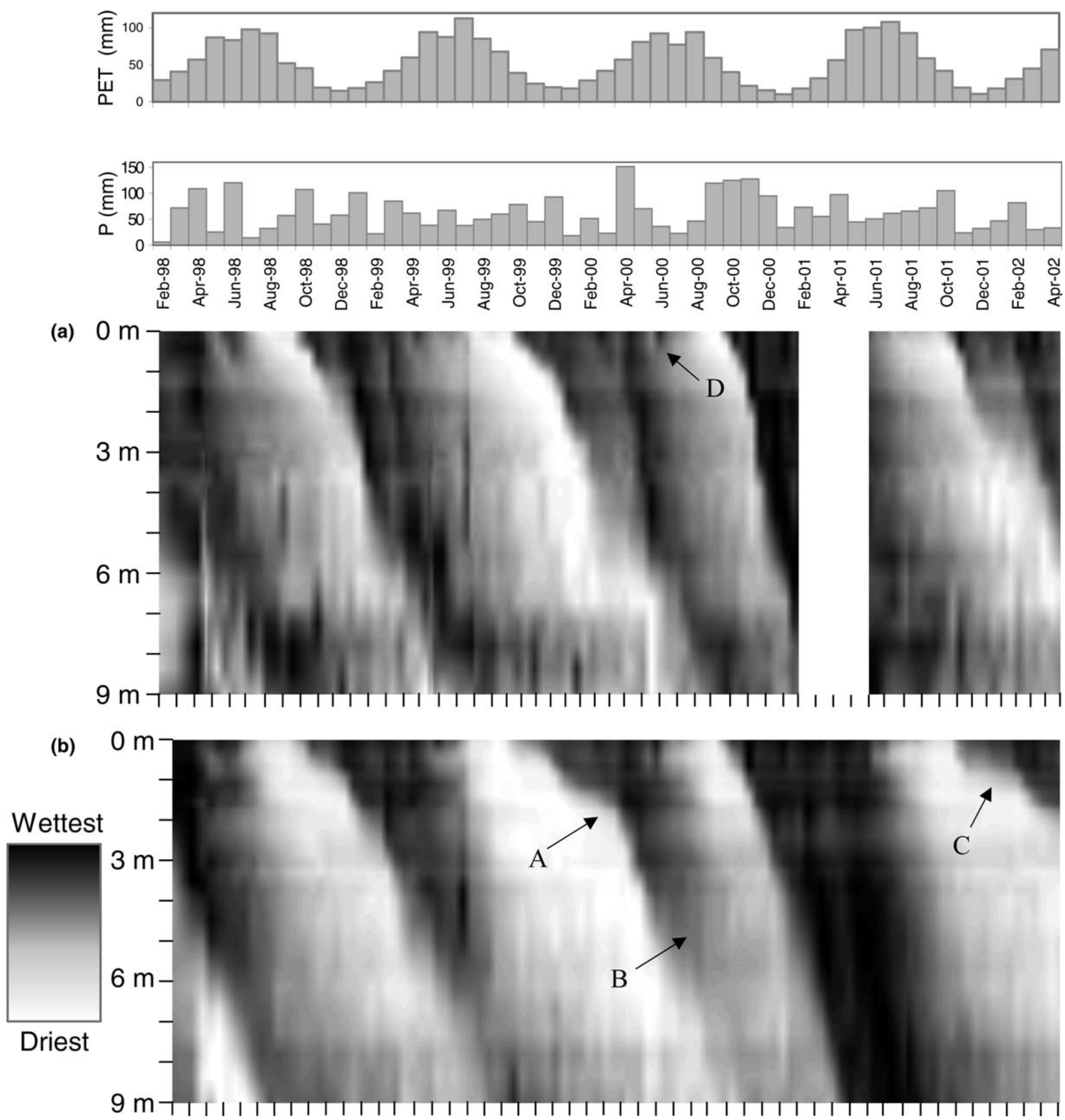

Figure 3 Relative moisture content of the 0—9 m soil—rock profile: (a) under grass ley from 12th February 1998 to 23rd April 2002; and (b) under oak woodland from 17th March 1998 to 23rd April 2002. See text for explanation of grey scale and annotations. Monthly rainfall (P) and potential evapotranspiration (PET) given above.

ative terms in Fig. 3. For each measurement depth and recording date, mean moisture values were derived for all neutron-probe access tubes at a site. These values were transformed into a 256-part grey scale, where black represents the maximum mean moisture content recorded at the specified depth during the period of observations, white represents the lowest mean moisture content recorded and intermediate values are assigned appropriate shades of grey. The grey-scale has then been used to create isopleths of relative wetness throughout the soil–rock column over the period of measurement using linear interpolation. In order to reduce the inherent noise in the resulting image, a down-profile-filter was applied to the raw data (each moisture value was substituted by the average value of itself, the layer above it and the layer below it).

This method of data portrayal is particularly useful in magnifying the subtle changes in water content that occur at depth in the profile, allowing visualization of processes that ordinarily go undetected or unnoticed because either the magnitude of their influence on moisture levels is small, or the vertical variation induced by differences in soil fabric masques their impact. Monthly rainfall and potential evapotranspiration are plotted above the soil moisture isopleth maps. Taken together, they provide a general guide to the likely direction of surface water flux at any point in the annual cycle.

Several patterns of moisture redistribution emerge. The most striking is the progression of the seasonal wetting front in autumn and early winter. At the grass site in years of near-normal autumnal rainfall (1998–1999, 1999–2000), rewetting of the profile to a depth of 5 m takes about 5–7 months. Under oak, this process is slower and occurs over approximately eight to nine months, reflecting both the effects of canopy interception in reducing net rainfall prior to leaf fall in the autumn and the need to recover the larger soil moisture deficit of the previous summer. The difference in behaviour between grass and oak may also reflect the potentially larger water-holding capacity of both the litter layer and the soil A horizon under oak with its greater organic matter content, resulting in less water being released for downward redistribution in the early stages of rewetting. There is no detectable difference between the sites in the rate of rewetting to 9 m, typically occurring between July and September, 10–12 months after rewetting of the topsoil, giving an average penetration rate for the wetting front to 9 m of 25–30 mm day^{-1} at both sites.

Progression of the wetting front is comparatively slow in the uppermost 2 m under oak. Label A in Fig. 3b highlights an inflection point, below which the descent of the wetting front accelerates. In some years, winter rains are insufficient to permit more than vestigial increases in moisture content below 5 m under oak (B, Fig. 3b).

The advantage of a comparatively long-term record of moisture content is that it reflects not only the normal seasonal changes in weather but also unusual events and patterns. Such was the case during Autumn 2001, which was exceptionally warm and dry. This led to delayed leaf-fall, the oak trees having retained about half their leaves even at the beginning of December, a pattern that compares with the normal half-defoliation date of mid November. Rainfall in November and December 2001 had been only 48% of the long-term mean, resulting in a much slower rewetting when compared with previous winters (Fig. 3). A combination of the reduced rainfall and prolonged interception by the oak canopy meant that the wetting front advance was stalled at around 0.9 m until the middle of February 2002 (C, Fig. 3b), only reaching 2.4 m by the end of the monitoring programme (April 2002), a depth that might ordinarily be passed in February during a year of normal weather. This compares with the grass site, where the penetration of the wetting front had reached 5 m by April 2002.

Another feature of the soil moisture regime of note is the seasonal progression of the drying front during spring and summer. In the topsoil, this process was temporarily reversed a number of times by heavy showers of rain, most noticeably in 1999 and 2000 under both land-uses (D, Fig. 3a). However, these occasional rewetting events were confined to the topsoil and moisture is not redistributed below 1 m at this time of year.

Previous research has identified the presence of preferential flow in sand soils (Ritsema and Dekker, 1996; Ritsema et al., 1997; Dekker et al., 2000; Hagedorn and Bundt, 2002). In order to gauge whether preferential flow is a significant feature at the sites in this study, the standard deviation of mean moisture change between successive measurement dates was determined using the equation:

$$\sigma_{\overline{\Delta\theta}} = \sqrt{\frac{\sum\left((\theta_i - \theta_{i-1}) - \overline{(\theta_i - \theta_{i-1})}\right)^2}{n-1}}, \tag{1}$$

where $\sigma_{\overline{\Delta\theta}}$ = standard deviation of the mean (of all tubes) moisture volume fraction change between successive measurement dates (dimensionless); n = number of measurement tubes at the site; θ_i = moisture volume fraction at a particular depth for an individual measurement tube on measurement occasion i.

If there were no preferential flow, access tubes would wet or dry in unison, producing low values of $\sigma_{\overline{\Delta\theta}}$. On the other hand, high values of $\sigma_{\overline{\Delta\theta}}$ within the four-year, 9-m profile matrix would highlight periods and depths when and where wetting or drying has been significantly uneven amongst the measurement tubes at a site. It is suggested that these high values of $\sigma_{\overline{\Delta\theta}}$ are diagnostic of the effects of preferential flow, especially where they lag progressively down-profile. These depth-temporal patterns are given in Fig. 4 for both sites using the same grey-scale interpolation technique deployed in Fig. 3, except that, in the case of Fig. 4, a down-profile-filter was not applied to the data.

Periods with high values of $\sigma_{\overline{\Delta\theta}}$ correspond with the leading edge of wetting fronts as they travel down the profile (A, Fig. 4b). This indicates that the soil wetted up at different rates at both sites, either because there was spatial variability in net rainfall and/or because there was preferential flow within the soil–rock profile. This pattern recurred in every year to 3 m and to at least 6 m in some years. Throughout the profile, the value of the standard deviation reduces following the occurrence of these spikes, suggesting that the spatial variability in wetting lasts for only about one month at any one depth and that, once moisture levels have started to recover from the summer minima, further moisture changes during winter are fairly even across each site.

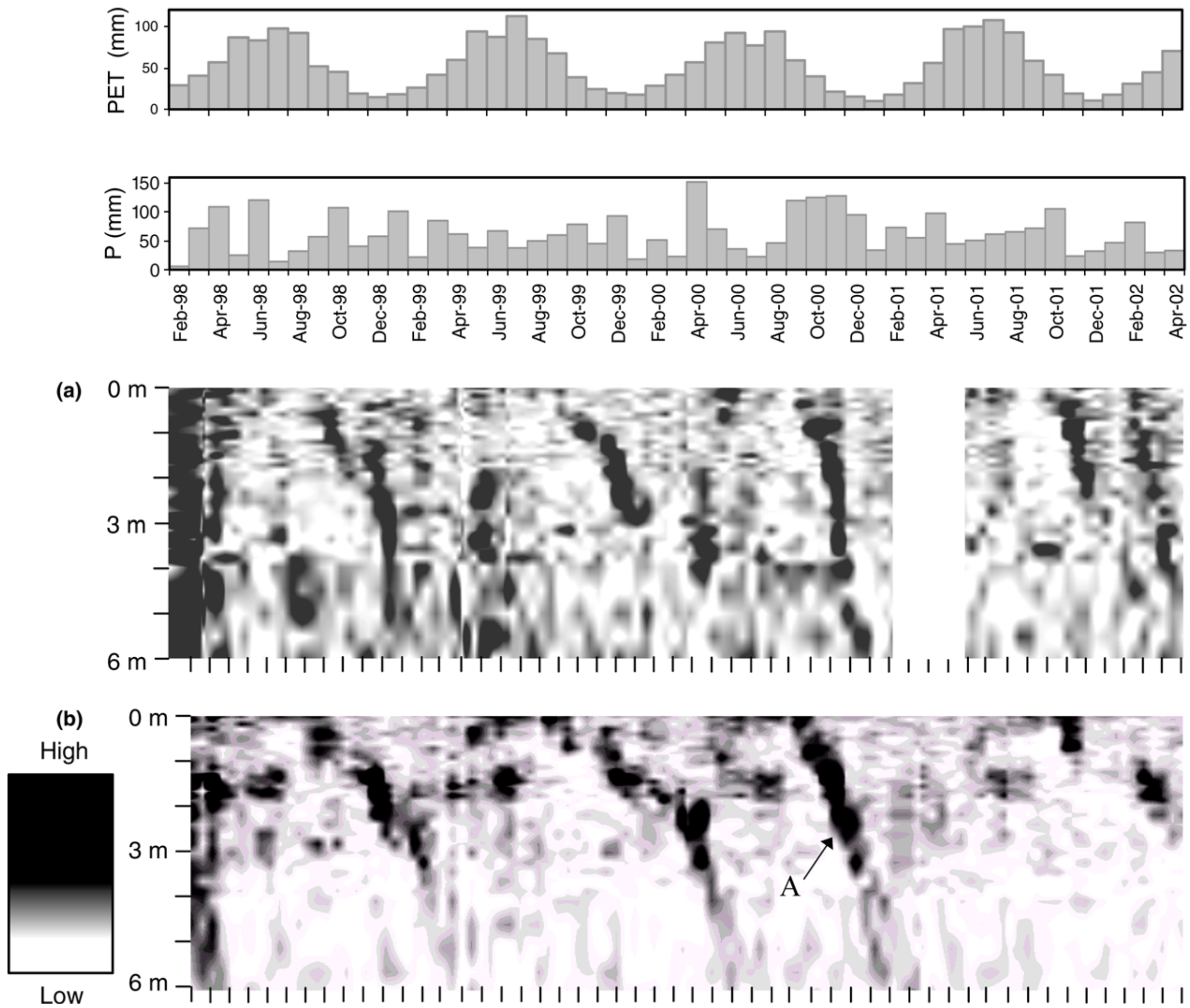

Figure 4 Standard deviations of the mean moisture content change between measurement dates, depth-by-depth: (a) throughout the 0–6 m profile under grass ley from 12th February 1998 to 23rd April 2002; and (b) throughout the 0–6 m profile under oak woodland from 17th March 1998 to 23rd April 2002. See text for explanation of the grey scale, the annotation and the missing data for the grass ley. Monthly rainfall (P) and potential evapotranspiration (PET) given above.

Soil moisture deficit

The water content of the uppermost 2 m of soil under each land cover for the period 12th February 1998–23rd April 2002 is given in Fig. 5. This depth holds all the plant roots. Both sites tend to return to plateau levels of moisture content during the winter periods. This is similar to other sandy soils (Ladekarl, 1998) and to a number of heavy clay soils (Reid and Parkinson, 1987; Calder et al., 1983). The behaviour contrasts with the loam soils overlying chalk where it has been suggested that an equilibrium winter condition equivalent to field capacity is never achieved (Wellings, 1984).

When compared with oak, the rapid decrease in soil moisture content under grass each spring is as expected, and is due to rapid sward development from as early as March and, hence, higher transpiration rates than from the ground flora beneath the defoliated oak. Indeed, the annual addition of manure or nitrogen fertilizer is intended to enhance this early growth in grass leys. However, the decline in soil moisture content appears to level out beneath grass from about the beginning of August. This suggests that there was water stress beneath this comparatively shallow-rooted sward, moisture volume fractions dropping below 0.035 within the uppermost 0.5 m in some years, compared with a permanent wilting point at the conventional pressure of -1.5 MPa of 0.065. There was also the effect of a reduction in transpiration following mowing. In contrast, appreciable drying of the soil under oak is delayed until June of each year, reflecting the timing of bud break in early May and subsequent leafing-out, which typically takes about 30 days (Bréda et al., 1993).

The lack of any indication that moisture values level out beneath oak each summer suggests that water was not limiting for trees at this site, at least under the ambient weather conditions of 1998–2002. The reduction in the rate of moisture depletion from mid-August may be attributable to the decline in potential evapotranspiration following

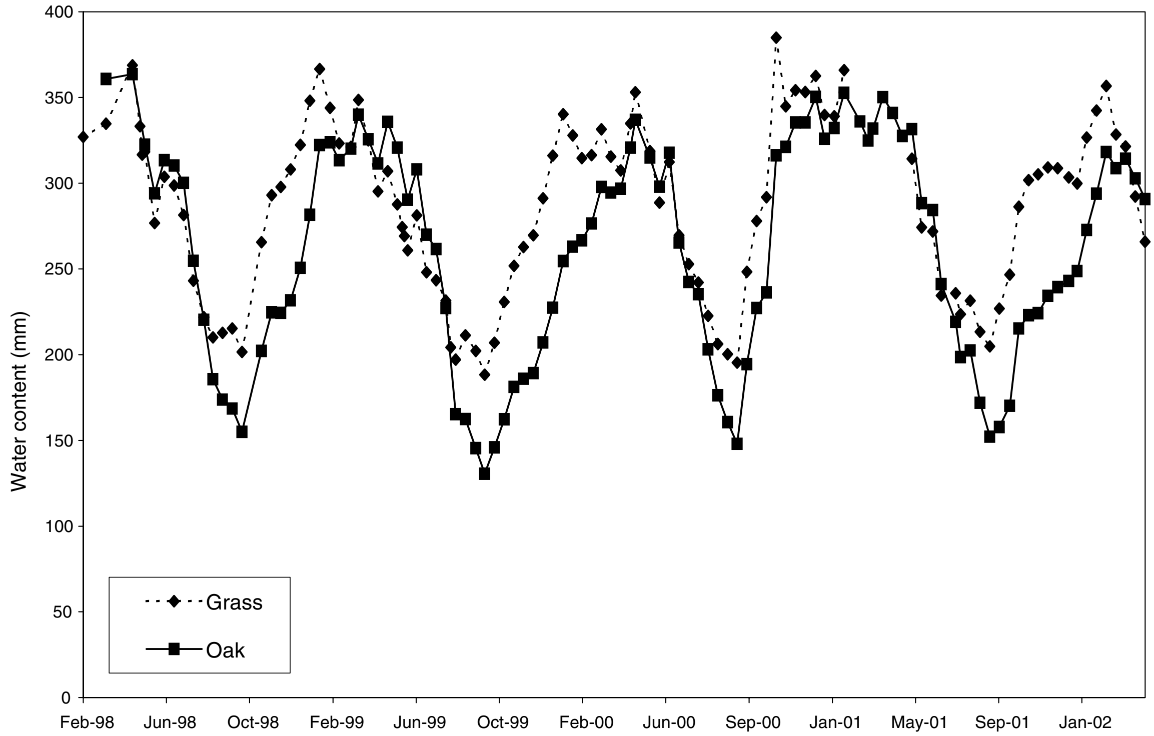

Figure 5 Four-year record (February 1998—April 2002) of mean water content of the uppermost 2 m of soil under grass ley and oak woodland as measured by neutron probe.

peak values in July. The lack of stress beneath oak contrasts with patterns established by Granier et al. (1999) for deciduous woodlands (including oak) at a number of sites in France, where soil moisture stress at the height of summer was shown to bring about stomatal closure and reduction in the ratio of actual/potential transpiration. The lack of any visible sign of water stress at Clipstone Forest may have been due to a number of factors, including the fact that the years during which field observation were made were comparatively wet so that soil moisture reserves were less depleted. Minimum moisture volume fractions do not drop below 0.07 within the uppermost 0.5 m and are never less than 0.065 in the upper 2 m soil block, while estimates of permanent wilting point at the conventional pressure of −1.5 MPa equate to a moisture volume fraction of 0.038 at this site. In addition, the site has a more northerly latitude and is more maritime in nature, both of which will mean that potential evaporative demand is lower than at sites in France.

Around August in each of the four years, the soil profile water content under oak dropped below that at the grass site. Also evident is a lag in the return to winter equilibrium levels of moisture beneath the oak site, typically at least one month behind grass (Fig. 5). This is partly due to the late leaf fall of this species and, hence, prolonged higher levels of interception in autumn and early winter. It also reflects the need to recover from greater summer soil moisture deficits — an extra 47—58 mm in each year when compared with the grass ley. Interesting here is that the seasonal lag is much less pronounced following the unusually heavy rainfalls of Autumn 2000 (Fig. 5).

Implications for groundwater recharge

The greater drying of the oak site and longer period away from winter equilibrium moisture conditions suggests that groundwater recharge will be less beneath this land-use than under the grass ley. Various estimates of recharge between March 1998 and March 2002 presented by Calder et al. (2002) indicate that recharge under oak was between 66% and 86% of that under grass. However, modelling of data from these sites (Calder et al., 2003) suggests that the average annual recharge over a 32.5-year period (1969—2002) was 136 mm for the grass site and 76 mm for the oak site, equating to recharge beneath oak being only 55% of that of grass.

The oaks in this study are 60 years old. Transpiration rates are thus likely to be close to their maximum and could decline to match those typical of short crops by the time the trees are 100 years old (Shiklomanov and Krestovsky, 1988). Consequently, the results given here are likely to illustrate the maximum divergence in water regime that might be expected between these two land covers. Differences in the magnitude of annual recharge beneath the two types of land-use will depend on stand density and age of the trees, as well as on grassland management.

Conclusions

The four-year record of moisture content in the sand soil-sandstone profiles at Clipstone Forest in the English midlands reveals both the cyclical patterns of moisture redistribution associated with the near-normal seasonal water balance of a temperate humid environment and the various impacts of unusual weather. The fact that the record extends to a depth of 9 m allows visualization of the seasonal downward progression of both wetting and drying in a way that is denied in most other studies.

Notable differences in the levels and incidences of seasonal soil moisture depletion and recovery are shown for grassland and oak woodland. Depletion is consistently larger under oak by approximately 50 mm. Recovery is generally faster under grass, resulting in the grass site being usually at least one month ahead of oak in its return to winter equilibrium moisture conditions. The greater drying and the absence of signs of water stress beneath the oak site accords with the ability of the trees to tap moisture from a greater depth than is possible by the shallower roots of the grass and to extract moisture at lower levels of moisture potential. However, since the ground beneath the oak site spends longer away from winter equilibrium moisture conditions, groundwater recharge here may be only 55% of that under the grass ley.

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